

GRID FORECAST ACCURACY UNDER STRESS

Eight U.S. balancing authorities, September 2024 to September 2026 | Jose Aleman | September 03, 2026

QUESTION

When the U.S. power grid is under the most stress, does the day-ahead demand forecast get worse exactly when accuracy matters most, and which regions are most exposed?

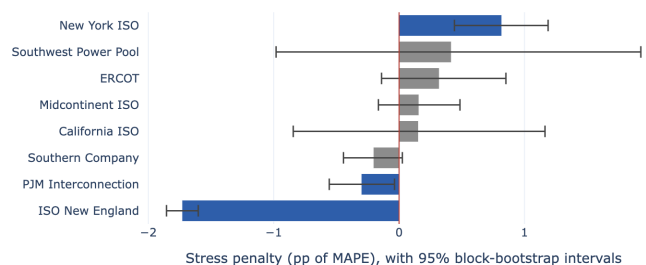
METHOD

Hourly demand, day-ahead forecast, net generation and interchange for eight balancing authorities (PJM, MISO, CISO, ERCOT, ISNE, NYIS, SWPP, SOCO) from the EIA-930 API: 139,055 balancing-authority-hours, 6,961 of them stress hours. A stress hour is the top 5% of demand within a balancing authority and season, which keeps differently sized grids comparable and stops the measure collapsing into summer versus winter. Python and DuckDB SQL. Intervals are 95% block bootstraps resampled by calendar day, since errors within a day are correlated and an hourly bootstrap would report noise as signal.

FINDINGS

1. The effect is real in one region, not eight. Pooled across all eight, forecast error does not worsen under stress (3.88% at stress vs. 3.93% normal), and with 95% block-bootstrap intervals attached only 3 of eight regions have a stress penalty distinguishable from zero. NYIS degrades (+0.82 pp, [+0.44, +1.19], a 1.31x multiple); ISNE and PJM improve, ISO New England sharply (-1.73 pp, 0.40x), consistent with operators watching days they already expect to be extreme. Everything else is noise: Southwest Power Pool has the second-largest point estimate at +0.42 pp but an interval of [-0.98, +1.93], entirely consistent with no effect. Ranking eight regions on point estimates alone would have put the wrong region first.
2. Against a naive baseline the forecasts hold up better than the raw error suggests. The day-ahead forecast beats 24-hour persistence (yesterday's demand at the same hour) by 26% at normal hours and 32% at stress hours: skill rises under stress, because persistence degrades faster than the forecast does. The exceptions are Southwest Power Pool and California ISO, both worse than persistence. They are not the same problem: SWPP's gap survives restricting to hours that balance, so it is real; CISO's does not.
3. Most of what looks like CISO forecast error is a reporting problem. CISO fails the EIA-930 balance identity in 6,396 of the 17,304 hours where the check can be run. On the hours that do balance its MAPE falls from 8.36% to 6.95% and its stress penalty flips from +0.15 to -1.67 pp. Ranking CISO among the worst forecasters measures its metering, not its forecasting.
4. The answer depends on what counts as stress. Defining a stress hour by net-load ramp instead of demand level picks out an almost disjoint set of hours (overlap is 5-12% by region) and reverses the ranking. Southern Company has the largest ramp-lens point estimate (+1.85 pp) and also the widest interval in the study, [+0.38, +4.56], so how big it is stays open. PJM Interconnection, which improves under demand stress, comes out slightly lower at +1.78 pp on an interval 8 times narrower, [+1.52, +2.04], a 1.52x multiple. Those are the only two ramp penalties that clear zero on the worsening side, and they disagree about size. Neither lens is wrong: one asks who is worst when load is highest, the other when it is moving fastest.

Only 3 of 8 regions differ from zero; NYIS is the only one that degrades under stress



RECOMMENDATION

Put stress-hour forecasting effort into New York ISO, and only New York ISO. It is the one region whose penalty survives a confidence interval (+0.82 pp, [+0.44, +1.19]), and it imports 2,399 MW at peak, so its misses land when its own headroom is thinnest. Spreading the same money across eight regions buys down an effect six of them do not measurably have. Two items should be funded separately, not as stress work: SWPP's day-ahead series loses to yesterday's demand in every hour, a baseline forecasting problem; and CISO's error is mostly a reporting problem. ERCOT is worth monitoring because it has only about 1.2 GW of DC-tie import capability against a ~85 GW peak, effectively isolated in stress conditions, but its penalty ([-0.14, +0.85]) is not established. For scale, the pooled stress-hour shortfall at an EIA-sourced \$40-200/MWh band is \$294M to \$1,468M over 24 months (\$734M central), an upper bound, not a measurement. One thing would change this answer: if the concern is ramp speed rather than demand level, PJM Interconnection displaces New York ISO on an effect that is larger and no less well established. Decide which kind of stress the budget is buying down before spending it.

LIMITATIONS

- MISO and SWPP span multiple time zones; each is assigned one local zone.
- Raw EIA series, not EIA's adjusted series, so reporting error stays visible rather than smoothed away.
- The price band is a sourced assumption, not a measurement, and the dollar figure assumes every under-forecast MW is bought at that price when reserves and demand response cover part of it: an upper bound.
- 24 months is short for weather-driven conclusions; a hot or mild year could shift the rankings.

SOURCE

